

Nov 3

Let x be a variable. For what x does $\sum_{n=0}^{\infty} \frac{x^n}{3^n}$ converge?

$$\frac{x}{3^n} = \left(\frac{x}{3}\right)^n \rightarrow \text{GS} \quad r = \frac{x}{3} \rightarrow |r| \text{ must be } < 1$$

↳ what if $|r| = 1$?

RT

$$a_{n+1} = \frac{1}{3 \cdot 3^n} \cdot x \cdot x^n = \frac{x \cdot x^n}{3 \cdot 3^n}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x \cdot x^n}{3 \cdot 3^n} \cdot \frac{3^n}{x^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x}{3} \right| < 1$$

↓
 $|x| < 3$

Must be < 1 ,
not $= 1$

$$(-3, 3) \rightarrow -3 < x < 3$$

If $x = 3$, $\sum_{n=0}^{\infty} \left(\frac{3}{3}\right)^n$ diverges by divergence test

If $x = -3$, $\sum_{n=0}^{\infty} \left(\frac{-3}{3}\right)^n = \sum_{n=0}^{\infty} (-1)^n$ diverges by divergence test